Time Varying Parameter Estimation: In Defense of Kernel Regression

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Background

Regression models of empirical phenomena conventionally assume model parameters are time-invariant. However, parameters may vary smoothly, drifting gradually over time, or break sharply in response to a discrete shock to the modeled phenomena. Outside of simulation exercises, the true data generating process is unknown, and thus the ex-ante assumption of fixed parameters is a strong one.

Econometric approaches to time-varying parameter estimation range from Markov regime switching models, which assume probabilistic transitions between multiple regimes in which parameters vary across regimes but are stable within a regime (Hamilton 1989), to structural break regression models (Chow 1960), which assume a single, discrete breakpoint in the empirical relationship that produces two sets of time-invariant parameters. Estimation generally involves formulating the model in state space and using the Kalman filter or smoother. The flexibility of representational forms for the time-varying parameter vector and the robust, widely available software suite for estimation make them natural choices. Rolling linear regression is another prevalent approach. Known derisively as a poor man's time-varying model (Zivot 2002), rolling OLS fits regression models to sliding windows of a fixed width and increment, and estimates constant parameters within the window. Rolling OLS is a form of Kernel Regression, which uses weighted least squares at each time index t to apply a kernel function that weights observations closes to t the most. Economic applications of kernel regression include return predictability tests and macro forecasting, but the principal upside of the method, aside from simplicity, is that it can incorporate any linear regression model without further assumptions.

I assess the performance of kernel regression[2] relative to state space modelling with simulation, specifically returns drawn from a time-varying CAPM calibrated with historical data. With a pre-specified data generating process, I can compare approaches by computing test error metrics directly instead of appealing to nebulous statistical properties of the resulting estimates. The study is not intended to be a comprehensive assessment of every candidate approach, but rather a demonstration that kernel regression can compete with state space models in a specific context.

[2] Kernel regression is the term used in the literature, but the approach can also be thought of as local regression. See the methodology section.

Setup

Adding a time subscript to the parameter vector β, yielding β_t, gives a standard linear model with time-varying parameters. For univariate y_t, x_t a kx1 vector of regressors, and β_t a vector of parameters:

$$y_t = x_t'\beta_t + \epsilon_t, t = 1...T$$

Assume covariates are exogenous and shocks are homoscedastic. This specification accommodates any regression model. To test whether kernel regression can adequately recover time-varying parameters, I use the CAPM with a time-varying factor loading and expected return[3] net of systematic factor exposure as a data generating process to simulate returns:

$$R_t - Rf_t = \alpha_t + \beta_t(Rm_t - Rf_t) + \epsilon_t, \epsilon_t \sim \mathrm{WN}(0, \sigma_\epsilon^2)$$
$$R_t^e = \alpha_t + \beta_t Rm_t^e + \epsilon_t$$

In lieu of exogenous processes, time series models underpin the dynamics of β_t, the ratio of the conditional covariance of the assets return with the market return to the variance of the market return, and the pricing error α_t. I assume parameters α_t and β_t are driftless random walks, to represent parameters with an expected change of zero. They drift randomly over time. All white noise terms are serially and mutually uncorrelated:

$$a_t = a_{t-1} + \eta_{1,t}$$
$$\beta_t = \beta_{t-1} + \eta_{2,t}$$

$$\begin{bmatrix} \epsilon_t \\ \eta_{1,t} \\ \eta_{2,t} \end{bmatrix} \sim \mathrm{WN}\left(\vec{0}, \Sigma_{\mathrm{WN}}\right)$$

$$\Sigma_{\mathrm{WN}} = \begin{bmatrix} \sigma_\epsilon^2 & 0 & 0 \\ 0 & \sigma_a^2 & 0 \\ 0 & 0 & \sigma_b^2 \end{bmatrix}$$

To simulate logarithmic market returns, I use a discretized geometric Brownian motion[4].

$$dS_t = uS_t dt + \sigma S_t dB_t^P$$
$$S_t = S_0 e^{(\mu - 0.5\sigma^2)t + \sigma B_t}$$
$$\ln\left(\frac{S_t}{S_0}\right) = (\mu - 0.5\sigma^2)t + \sigma B_t$$
$$R_t \equiv \ln(S_t/S_{t-1})$$
$$= (\mu - 0.5\sigma^2) + \sigma W, W \sim N(0, 1)$$

Algorithmically, the simulation exercise proceeds as follows:

1. Simulate N=1,000 length T=2,500 draws of Rm_t-Rf_t, α_t, β_t, and ε_t

2. Construct response variable R_t-Rf_t

3. For each draw i=1...N,

    A. estimate $\widehat{\alpha}_{it}$, $\widehat{\beta}_{it}$ with kernel regression and a state space model

[3] See the Appendix for a summary of rolling OLS and a derivation of the CAPM warranting time variation in factor loadings.

[4] A GBM is a continuous time stochastic process, but simulation parameter processes are discrete. In practice, simulating a GBM involves discrete, standard normal draws scaled by small time intervals. See the Appendix for the Euler discretization.

B. Compute R^2, mean-squared error, and mean absolute percentage error between estimates and simulated α_{it}, β_{it}

The table below contains simulation parameters[5].

Simulation Parameter Values

Parameter	Estimate	Calibration Details
$\hat{\mu}_{GBM}$	0.0005326	Daily SP500 log returns, 5/18/2025 – 5/14/2025 (T=2420)
$\hat{\sigma}^2_{GBM}$	0.0001359	
$\hat{\sigma}_a$	0.0001	Sample standard deviation of AR(1) regression of T=2148 OLS parameter estimates, parameter estimates obtained using rolling OLS on T=2401 daily AAPL and SP500 excess log returns (5/18/2015 – 5/13/2025) with W=253.
$\hat{\sigma}_b$	0.0177	
$\hat{\sigma}^2_e$	0.00034	Sample variance of CAPM OLS on T=2401 daily AAPL and SP500 excess log returns (5/18/2015 – 5/13/2025)
S_0	2122.73	SP500 closing price on 5/15/2015.

Simulation parameters. T=2500, N=1000, using 20% burn in and multivariate normal draws. Initial values are random unless otherwise stated.

[5] Several parameters for the simulation are estimates from rolling OLS. For comparison of kernel regression to state space models, this ostensibly is problematic. However, the first stage rolling OLS estimates only generate estimates of the error term's variance, as simulation assumes a random walk process. Variance and volatility estimates are quite small, and should be in the context of daily return simulation. Higher beta variance accords with economic intuition. If the time-varying CAPM holds, alpha should be near zero in magnitude and beta should exhibit more variance because the stock return's co-movement with the market factor is more likely to change over time than systematic returns net of exposure to the market factor, which should average zero. Moreover, I test the parameter recovery of kernel regression and state space models on the same data. One set of simulated parameters' volatility doesn't bear on a *relative* comparison of performance.

Methods

Summary of Approaches

Estimation Method	Details
Kernel Regression	Gaussian Kernel, $h = T^{0.5}$
State Space Model (KF)	0 for initial state mean and I_2 for initial state covariance matrices

Kernel Regression

I outline kernel regression. First, a valid Kernel function is non-negative, symmetric, and integrates to 1 over its support:

$$1. K(x) \geq 0 \; \forall x$$
$$2. \int_{-\infty}^{\infty} K(u)\mathrm{du} = 1$$
$$3. K(x) = K(-x) \; \forall x$$

For example, rolling OLS uses a boxcar (alternatively, a uniform or rectangular) kernel:

$$K(x) = \mathbf{1}(|x| \leq c) \cdot \frac{1}{2c}$$

The Gaussian kernel has the real line as a support:

$$K(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}}$$

Unlike the boxcar kernel, all observations get a strictly positive weight, but the weight decays rapidly for large $|x|$. The kernel function $K((t-i)h^{-1})$ can assign greater weight to observations close to t and less weight to more distant observations. Smoothing parameter h is the bandwidth. Larger h results in smoother estimates, smaller h, noisy estimates. A boxcar kernel can induce more variance in rolling estimates because of the sharp transition (from constant to 0 at the boundary). The bandwidth h can be calibrated using cross validation or can be tied to the sample size[6]. I use a Gaussian kernel with $h=T^{0.5}$.

Kernel regression performs weighted least squares at each time index to generate time varying parameter estimates. Let W_t be a diagonal matrix with time-varying elements $K((t-1)h^{-1})$, $K((t-2)h^{-1})$, ..., $K((t-(T-1))h^{-1})$, $K((t-T)h^{-1})$, such that $W_t = W_t'$, X a T x 2 design matrix, and Y a T x 1 vector of responses:

$$W_t = \begin{bmatrix} K(\frac{t-1}{h}) & \cdots & & 0 \\ & K(\frac{t-2}{h}) & & \\ \vdots & & \ddots & \vdots \\ & & K(\frac{t-(T-1)}{h}) & \\ 0 & & & K(\frac{t-T}{h}) \end{bmatrix}_{T\times T}$$

$$X = \begin{bmatrix} 1 & x_1 \\ \vdots & \\ 1 & x_T \end{bmatrix}_{T\times 2}, \; Y = \begin{bmatrix} y_1 \\ \vdots \\ y_T \end{bmatrix}_{T\times 1}, \; \beta_t = \begin{bmatrix} \beta_{0,t} \\ \beta_{1,t} \end{bmatrix}_{2\times 1}$$

$$Y = X\beta_t + \epsilon, \epsilon \sim N(\vec{0}, \sigma_\epsilon^2 I_{TxT})$$

[6] See the Appendix for candidate CV procedures. Owing to computational cost, I use don't pursue these further.

Covariate and response matrices are time-invariant. At each t=1...T, parameter estimates, standard errors[7], confidence intervals, and p-values follow from:

$$\begin{aligned}
\hat{\beta}_t &= (X'W_tX)^{-1}(X'W_tY) \\
\mathrm{Var}\left(\hat{\beta}_t|X\right) &= (X'W_tX)^{-1}\mathrm{Var}\left(X'W_tX\beta_t + X'W_t\epsilon|X\right)(X'W_tX)^{-1} \\
&= (X'W_tX)^{-1}X'W_t\mathrm{Var}(\epsilon|X)W_t'X(X'W_tX)^{-1} \\
&= (X'W_tX)^{-1}X'W_t\sigma_e^2 I W_t'X(X'W_tX)^{-1} \\
&= \sigma_e^2(X'W_tX)^{-1}X'W_tW_t'X(X'W_tX)^{-1} \\
&= \sigma_e^2(X'W_tX)^{-1}X'W_t'W_tX(X'W_tX)^{-1}
\end{aligned}$$

Note, $X'W_tX$ is symmetric, and $(A')^{-1} = (A^{-1})'$ generally, so $((X'W_tX)')^{-1} = (X'W_tX)^{-1}$. The estimate of variance σ^2_e uses the weighted sum of squared errors, and is also time-varying:

$$\hat{\sigma}^2_{e,t} = \frac{(Y - X\hat{\beta}_t)'W(Y - X\hat{\beta}_t)}{T-2-1}$$

$$\hat{\mathrm{Cov}}(\beta_t) = \hat{\sigma}^2_{e,t}(X'W_tX)^{-1}X'W_t'W_tX(X'W_tX)^{-1}$$

Standard errors are the square root of the diagonal elements of the estimated covariance matrix, with corresponding 95% confidence intervals:

$$\hat{\beta}_{0,t} \pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_{0,t})$$
$$\hat{\beta}_{1,t} \pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_{1,t})$$

Performing kernel regression at each t produces parameter estimates and therefore in sample predictions, with associated MAPE and R^2, that vary over time. Prediction quality of observation t should be highest given the kernel weights the observations at t the most, so point APE at t is a key metric:

$$\mathrm{MAPE}_t = \left|\frac{X\hat{\beta}_t - Y}{Y}\right|$$

$$\mathrm{APE}_t = \left|\frac{x_t\hat{\beta}_t - y_t}{y_t}\right|$$

State Space Model

Before applying the Kalman filter[8], I first recast the time-varying CAPM model:

$$\begin{aligned}
R_t - Rf_t &= \alpha_t + \beta_t(Rm_t - Rf_t) + \epsilon_t \\
R_t^e &= \alpha_t + \beta_t Rm_t^e + \epsilon_t \\
a_t &= a_{t-1} + \eta_{1,t} \\
b_t &= b_{t-1} + \eta_{2,t}
\end{aligned}$$

in state space form. Define the state and measurement equations as follows:

$$\begin{aligned}
X_t &= \phi X_{t-1} + W_t \\
Y_t &= A_t X_t + V_t
\end{aligned}$$

[7] The notation assumes homoscedastic errors: $\mathrm{Var}(e|X) = \sigma^2_e I$. The derivations carry through with a more complicated structure.

[8] A Kalman filter provides filtered parameter estimates at t using observations up to t. A Kalman smoother runs the filter forward through the time series and then performs a backward pass, thus leveraging all data. To estimate parameters over a historical (albeit simulated) dataset, a smoother is a better point of comparison to a kernel regression, which at any index t_0 assigns non-negative weight to all observations, including those at $t>t_0$. I use a filter for several reasons. First, at T=2,500 and N=1,000, the computational cost of a smoother is prohibitive. Second, a filter provides online estimates of the parameters as new data arrives sequentially, making it more applicable for forecasting. Third, simulated time varying parameters are random walks, covariates are IID draws, and noise is IID, hence y_{t+1} shouldn't bear on estimates of a_t and b_t.

Y_t is observed, A_t the observation matrix, X_t the unobserved state variable, and W_t and V_t are noise terms:

$$X_t \equiv \begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix}, \Phi = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, W_t = \begin{bmatrix} \eta_{1t} \\ \eta_{2t} \end{bmatrix}$$

$$Y_t = R_t^e, A_t = [1 \ Rm_t^e], V_t \equiv \epsilon_t$$

Further, let:

$$\begin{aligned} W_t &\overset{\text{iid}}{\sim} N(0, Q) \\ Q &= \begin{bmatrix} \sigma_a^2 & 0 \\ 0 & \sigma_b^2 \end{bmatrix} \\ V_t &\overset{\text{iid}}{\sim} N(0, R) \\ R &= \sigma_e^2 \end{aligned}$$

The Kalman filter is a projection:

$$X_t^t \equiv E(X_t | Y_1 \ldots Y_t)$$

with corresponding conditional error variance:

$$P_t^t = E[(X_t - X_t^t)(X_t - X_t^t)' | Y_1 \ldots . Y_t)$$

Set two initial conditions

$$\begin{aligned} X_0^0 &= x_0^0 \\ P_0^0 &= \Sigma_0 \end{aligned}$$

and for t=1...T, iteratively compute the state update equations, Kalman filter, MSE, and Kalman gain:

$$\begin{aligned} X_t^{t-1} &= \Phi X_{t-1}^{t-1} \\ P_t^{t-1} &= \Phi P_{t-1}^{t-1} \Phi' + Q \\ X_t^t &= X_t^{t-1} + K_t(Y_t - A_t X_t^{t-1}) \\ P_t^t &= [I - K_t A_t] P_t^{t-1} \\ K_t &= P_t^{t-1} A_t' [A_t P_t^{t-1} A_t' + R]^{-1} \end{aligned}$$

Once formulated as a state space model, estimation is straightforward with MLE.

Results

I present summary statistics, computed over N=1000 simulations, for the errors between estimates of R_t^e, a_t, and b_t obtained using kernel regression and the state space model, and simulated actuals. Each simulation contains T=2500 known values of the parameters, and applying each method to the simulated R_t^e produces T=2500 estimates of a_t and b_t.

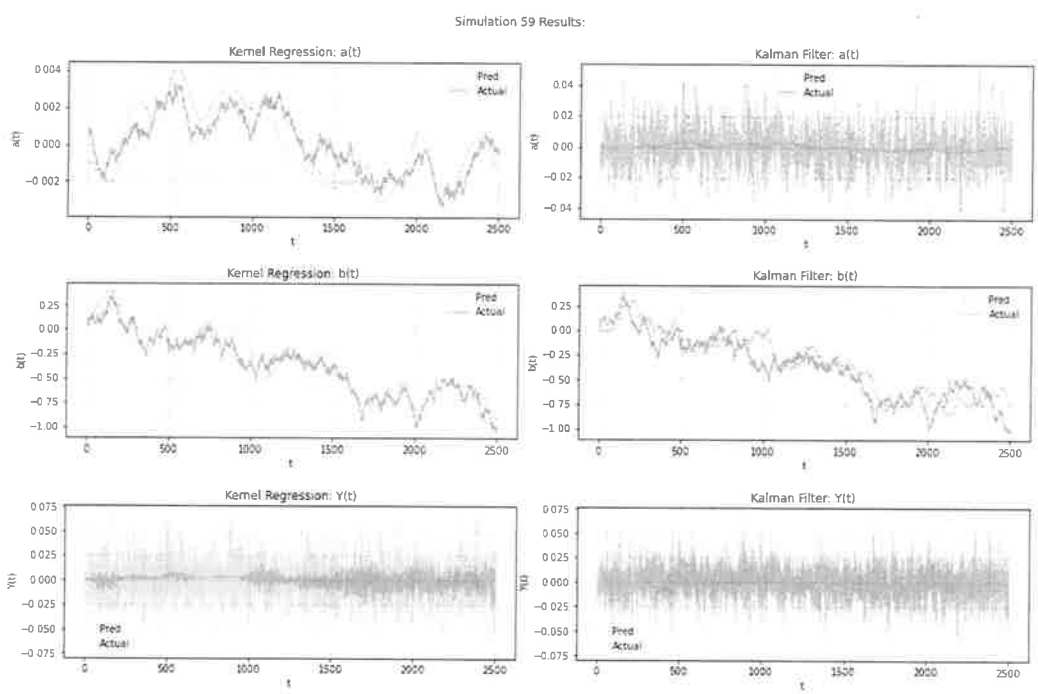
Kernel Regression R^2

	Mean	Std.	Min.	P25	Med.	P75	Max.
a_t	0.174	0.666	-4.696	-0.072	0.370	0.637	0.945
b_t	0.745	0.217	-0.475	0.668	0.808	0.893	0.981
R_t^e	0.180	0.129	0.025	0.082	0.138	0.243	0.729

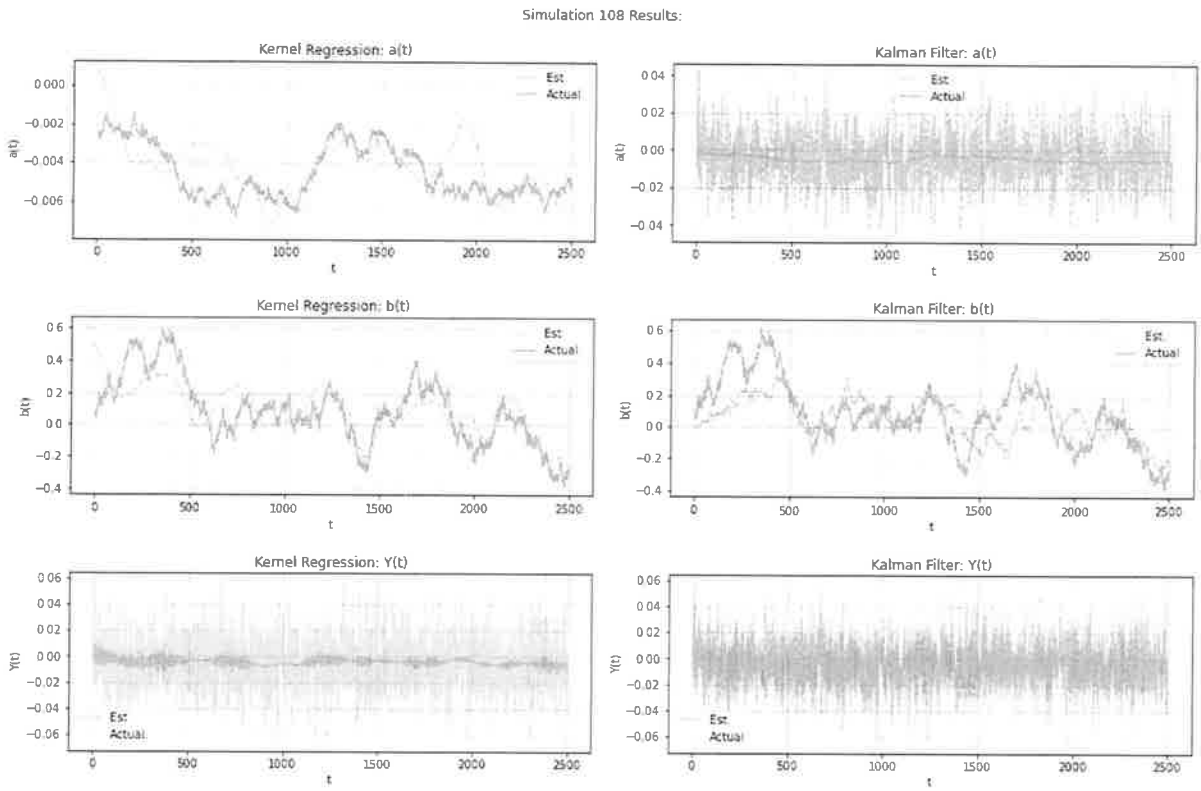
State Space R^2

	Mean	Std.	Min.	P25	Med.	P75	Max.
a_t	-58.958	44.142	-338.343	-77.542	-46.109	-26.422	-4.225
b_t	0.511	0.392	-2.452	0.378	0.612	0.770	0.954
R_t^e	0.824	0.027	0.790	0.804	0.815	0.837	0.941

The state space model outperforms kernel regression handily for R_t^e. Median R^2 is 81.5%, compared to kernel regression's 13.8%. However, kernel regression better estimates the time-varying factor loading, b_t, by every summary statistic. The average and median R^2 is 74.5% and 80.8%, respectively, compared to the state space model's 51.1% and 61.2%. For the expected excess return, a_t, the state space model performs disastrously. Across all simulations, the Kalman filter's R^2 for a_t is negative, indicating considerably worse performance than predicting the in sample mean of zero. To understand how the state space model can simultaneously produce higher R^2 for R_t^e, but lower R^2 for a_t and b_t compared to Kernel regression, I plot estimated versus actual a_t, b_t, and R_t^e for an arbitrary simulation.



Mechanically, it is a result of much greater variability[9] in state space estimated a_t, which translates to much greater variability in predicted R_t^e and therefore higher R^2. Kernel regression's fit for R_t^e cannot explain as much of the variation in actual R_t^e because the fit's variance is considerably lower than that of actual. Examining the parameter a_t and b_t error distributions shows kernel regression's superior performance, especially when the process for a time-varying parameter, e.g. a_t, exhibits little variance.



The above plots show in sample estimates for another simulation. Kernel regression better captures the time-varying factor loading b_t and the state space model's estimates of a_t drastically overstate variance.

[9] The variability in the state space estimated a_t stems from the initial conditional error variance, set to 1. The actual sample variance of residuals from the AR(1) regression of rolling OLS's a_t estimates is effectively 0, highlighting the sensitivity of the state space model's fit to initial values.

Conclusion

Linear models with time-varying parameters capture dynamic relationships between covariates and the response variable. Once posited in state-space form, in which the time-varying parameters are latent states that evolve over time per a defined process, such models are often estimated using the Kalman filter or smoother. State-space models offer several advantages. First, they explicitly detail the evolution of the parameters with assumed dynamics, such as a random walk or local trend. Second, if the assumptions of linearity and normally distributed noise hold, the Kalman filter is the mean-square error minimizing linear estimator. Third, the filter is computationally efficient and conducive to forecasting because it uses recursive estimation, updating estimates as new data arrives. Disadvantages include sensitivity to initial state values and crucially the assumed time series dynamics of the parameters. A simulation exercise renders the latter concern moot, but with actual data, biased estimates from mis-specified dynamics is an acute concern. Parameters may exhibit structural breaks, for example, instead of drifting slowly per a random walk.

Kernel regression performs local weighted least squares at each time t, applying weights determined by a kernel with observations temporally closer to t receiving higher weights, and farther observations receiving lower weights. The primary advantage is non-parametric estimation. Parameters can drift or change in a nonlinear manner; kernel regression doesn't assume functional forms for parameter dynamics. The tradeoff is that parameter estimates are affected greatly by local data, including aberrational values. Although I used a common heuristic for kernel regression's bandwidth, a key hyperparameter, in practice optimal bandwidth selection is an additional concern. Additional disadvantages are biased estimates at the beginning and end of the time series, less interpretability owing to a lack of a specified form for parameter evolution, and complications with forecasting.

Using a large sample (T=2,500) of daily stock returns simulated from the CAPM with time-varying excess expected return and market factor loading parameters, I show that kernel regression can outperform a state space model and therefore a state space model shouldn't be the default approach. I conclude the following. First, the choice of estimation method is downstream from the supposition of the true underlying behavior of the time-varying parameters. If there is decisive rationale to assume the parameter dynamics, a state space model is appropriate. Absent such rationale, kernel regression is flexible, non-parametric, and permits the local data to drive the parameter dynamics. Second, both methods can produce biased parameter estimates. The strength of the bandwidth selection procedure for kernel regression should be compared to the strength of the state space model's assumption for the parameters dynamics. The former, which often involves cross-validation, again leverages the data. The latter leverages apriori domain knowledge. Third, the intended application can be decisive. For generating interpretable forecasts, a state space model is a natural approach. For pure estimation, such as in the context of historical analysis, kernel regression may be more appropriate.

In the following sections, I perform additional simulation testing to isolate the effect of a two-sided window on kernel regression's performance, and detail more nuanced versions of kernel regression.

Underview: Additional Testing

A kernel regression at t accesses all data, assigning high but decreasing weights to observations t+1, t+2, etc. If observations at $t>t_0$ enhance the quality of estimates at t_0, kernel regression has a natural advantage over a Kalman filter. To assess the benefit of a two-sided filter for kernel regression, I perform additional testing on separate simulated series using a one-sided Gaussian filter and the ACF derived window function. Again, I use random walks for the parameters and IID normal draws for the covariate and shocks:

$$Y_t = a_t + b_t X_t + \epsilon_t$$
$$a_t = \mu_a + \phi_a a_{t-1} + w_{t,a}$$
$$b_t = \mu_b + \phi_b b_{t-1} + w_{t,b}$$
$$[X_t \; e_t \; w_{t,a} \; w_{t,b}]' \overset{\text{i.i.d.}}{\sim} \text{MVN}(\mathbf{0}, I_4)$$

I use a bandwidth of $h=\sqrt{T}$. To obtain ϕ for the ACF window function, I perform a whole sample AR(1) regression and use the absolute value of the estimated ϕ. The one-sided Gaussian window function simply truncates the Gaussian kernel at $x > 0$:

$$\text{Gaussian: } K(x) = Ce^{-0.5x^2}$$
$$\text{One-Sided: } K(x) = \mathbf{1}\{x \leq 0\} \cdot Ce^{-0.5x^2}$$
$$\text{ACF: } K(x) = \mathbf{1}\{x \leq 0\} \cdot |\widehat{\phi}|^x$$

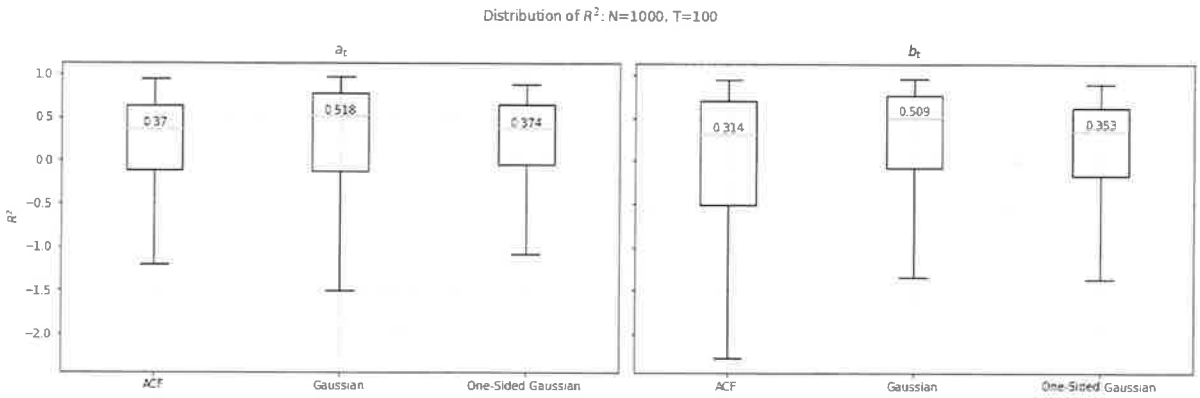
I simulate N = 1000 paths of T=150 observations from the above process, using parameters, ϕ_a =1.0, ϕ_b =1.0, μ_a = 0.0, μ_b = 0.0, and compute error measures on the last 100 simulated observations to remove biased error measures (owing to the small sample size) from one-sided windows observed at t<50. For each simulated series, I compare estimated a_t and b_t to actual, compute R^2, and report the distributions:

Distribution of R^2, ϕ_Y

Window Function	ACF		Gaussian		One-Sided		ϕ_Y: $Y_t = B_0 + \phi_Y Y_{t-1} + u_t$
	a_t	b_t	a_t	b_t	a_t	b_t	
N	1000						
Mean	0.024	-0.689	-0.075	-0.085	0.165	0.097	0.269
Std.	1.149	4.441	1.875	1.913	0.746	0.789	0.217
Min.	-12.760	-94.273	-27.778	-18.956	-5.280	-5.409	-0.071
1%	-4.515	-13.380	-7.506	-10.212	-2.897	-3.115	-0.054
25%	-0.110	-0.509	-0.135	-0.077	-0.049	-0.173	0.093
50%	0.370	0.314	0.518	0.509	0.374	0.353	0.232
75%	0.640	0.699	0.787	0.771	0.653	0.628	0.415
99%	0.892	0.926	0.950	0.939	0.857	0.872	0.821
Max.	0.947	0.949	0.975	0.971	0.884	0.899	0.943

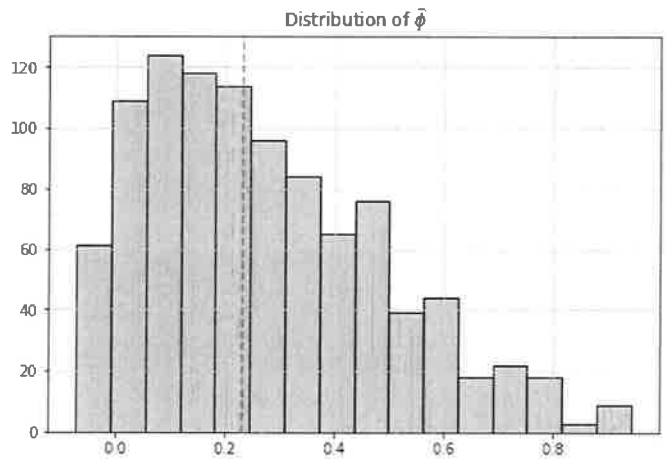
Simulation results using N=1000, T=100, and driftless random walks for a(t) and b(t). Apply kernel regression, using h=T^0.5, with a Gaussian kernel, one-sided Gaussian window, and ACF window. Calibrate the ACF window's ϕ by regressing y(t) on y(t-1) using all T=150 observations.

Median R^2 for a_t and b_t using a Gaussian kernel is 51.8% and 50.9%, compared to 37.4% and 35.3% for a one-sided Gaussian window. The reductions of roughly 14-15% in R^2 are modest and similar to those observed for the ACF window function. The ACF and one-sided Gaussian window functions perform similarly, although the former's performance exhibits more variability.



Box and whiskers plot of simulation R^2 values per kernel/window function and time-varying parameter.

Owing to the random walk parameters, and IID draws for the covariate and noise, the simulated process Y_t exhibits low persistence. I plot the distribution of whole sample AR(1) estimates below. Estimated ϕ values are centered around 0.2.



Despite this, the results show that kernel regression benefits noticeably from a two-sided window function. Some loss in explanatory power is to be expected when using a one-sided window function, even when there is no reason ex-ante to expect that observations at $t > t_0$ bear on estimation of parameters at t_0, and the simulation results provide a data point as to how much (e.g. 14-15% R^2). For context, the kernel regression achieves 20% higher median R^2 than the state space model in the primary simulation exercise.

Appendix

CAPM Derivation

In this section, I outline the case for time-varying parameters by deriving the conditional CAPM. In expected return-beta form, the unconditional CAPM is:

$$E(R^i - R^f) = \beta_{i,R^w}\left[E(R^W) - R^f\right]$$
$$\beta_{i,R^w} = \frac{\mathrm{Cov}(R^i, R^W)}{\mathrm{Var}(R^W)}$$

for return R^i, R^w the return on the total wealth portfolio, often proxied using the return on a stock index, R^f the risk-free rate, and β_{i,R^w} the regression coefficient of R^i on R^w. An asset's expected excess return equals β_{i,R^w}, the amount of asset i's exposure to the market factor's risk, multiplied by the factor risk premium, or equivalently the market price of risk, $\lambda_W = E(R^W) - R^f$.

Thus far, β_{i,R^w} is the ratio of the unconditional covariance of returns to the unconditional variance of the wealth return. Empirical tests of the CAPM, and factor models broadly, involve a panel of asset returns, and first performing a time series regression for each asset's excess returns to estimate β_{i,R^w}:

$$R_t^i - R^f = \alpha_i + \beta_i(R_t^m - R^f) + \epsilon_t^i$$

followed by a cross-sectional regression of excess return on beta per time period, to estimate λ_W. If the model's hypothesis is true, α_i, the regression intercept and pricing error, should be statistically zero.

Empirical tests of the CAPM and other factor models often assume time-invariant factor loadings as a consequence of CAPM using unconditional distributional moments. Conditional moments of asset returns – expectations, variances, and covariances - condition on the available information set at time t, e.g. $E_t(R_{t+1}) = E(R_{t+1} | I_t)$, and are likely not equal to unconditional moments because returns are not IID. Note, the CAPM can hold conditionally, meaning be true in the cross-section for any period t, but fail unconditionally over multiple periods. I turn to the motivation for time-varying parameters.

Begin with the fundamental asset pricing equation from an agent's two period consumption problem (note, the result follows for any two periods of a multiperiod model):

$$p_t = E_t\left[\beta\frac{u'(C_{t+1})}{u'(C_t)}x_{t+1}\right]$$

Equivalently:

$$p_t = E\left[\beta\frac{u'(C_{t+1})}{u'(C_t)}x_{t+1}\middle| I_t\right]$$

Let the SDF and gross return be, respectively:

$$m_{t+1} \equiv \beta\frac{u'(C_{t+1})}{u'(C_t)}$$
$$R_{t+1} \equiv \frac{x_{t+1}}{p_t}$$

Using the covariance decomposition E(XY) = E(X)E(Y) + Cov(X,Y):

$$1 = E_t(m_{t+1}R_{t+1})$$
$$1 = E_t(m_{t+1})E_t(R_{t+1}) + \mathrm{Cov}_t(m_{t+1}, R_{t+1})$$

Normalize the price of the risk-free asset to 1, and denote its gross return as R^f:

$$1 = E_t\left[m_{t+1} R^f\right]$$
$$R^f = \frac{1}{E_t(m_{t+1})}$$

The CAPM assumes the SDF is linear in the return on a total wealth portfolio, where a_t and b_t are free parameters:

$$m_{t+1} = a_t + b_t R^w_{t+1}$$

Price the risk-free asset and the wealth portfolio, solving for a_t and b_t and adding a time subscript to the risk-free rate to make the time dependence and relevance of the information set explicit:

$$\begin{aligned} 1 &= E_t(m_{t+1} R_{t+1}) \\ 1 &= E_t(m_{t+1} R^f_t) \\ &= R^f_t E_t(m_{t+1}) \\ 1 &= E_t(m_{t+1} R^W_{t+1}) \end{aligned}$$

I obtain a_t:

$$\begin{aligned} 1 &= E_t(m_{t+1} R^f_t) \\ \frac{1}{R^f_t} &= E_t(a_t + b_t R^W_{t+1}) \\ a_t &= \frac{1}{R^f_t} - b_t E_t(R^W_{t+1}) \end{aligned}$$

And solve for b_t:

$$\begin{aligned} 1 &= E_t(m_{t+1}) E_t(R^W_{t+1}) + \mathrm{Cov}_t(m_{t+1}, R^W_{t+1}) \\ 1 &= \frac{1}{R^f_t} E_t(R^W_{t+1}) + \mathrm{Cov}_t(a_t + b_t R^W_{t+1}, R^W_{t+1}) \\ 1 - \frac{E_t(R^W_{t+1})}{R^f_t} &= b_t \mathrm{Var}_t(R^W_{t+1}) \\ b_t &= \frac{R^f_t - E_t(R^W_{t+1})}{R^f_t \mathrm{Var}_t(R^W_{t+1})} \\ &= \frac{E_t(R^W_{t+1}) - R^f_t}{-R^f_t \mathrm{Var}_t(R^W_{t+1})} \end{aligned}$$

With a_t and b_t, I derive the conditional CAPM:

$$\begin{aligned} 1 &= E_t(m_{t+1} R_{t+1}) \\ 1 &= E_t(m_{t+1}) E_t(R_{t+1}) + \mathrm{Cov}_t(m_{t+1}, R_{t+1}) \\ 1 &= \frac{E_t(R_{t+1})}{R^f_t} + \mathrm{Cov}_t(m_{t+1}, R_{t+1}) \\ E_t(R_{t+1}) - R^f_t &= -R^f_t \mathrm{Cov}_t(m_{t+1}, R_{t+1}) \\ E_t(R_{t+1}) - R^f_t &= -R^f_t \mathrm{Cov}_t(a_t + b_t R^W_{t+1}, R_{t+1}) \\ &= -R^f_t \frac{E_t(R^W_{t+1}) - R^f_t}{-R^f_t \mathrm{Var}_t(R^W_{t+1})} \mathrm{Cov}_t(R^W_{t+1}, R_{t+1}) \\ &= \frac{\mathrm{Cov}_t(R^W_{t+1}, R_{t+1})}{\mathrm{Var}_t(R^W_{t+1})} (E_t(R^W_{t+1}) - R^f_t) \\ &= \beta_t \left[E_t(R^W_{t+1}) - R^f_t\right] \end{aligned}$$

and use the LIE to condition down:

$$\lambda_t \equiv E_t(R^W_{t+1}) - R^f_t$$
$$E(E_t(R_{t+1} - R^f_t)) = E(\beta_t \lambda_t)$$
$$E(R_{t+1} - R^f_t) = E(\beta_t)E(\lambda_t) + \mathrm{Cov}(\beta_t, \lambda_t)$$

If the covariance between the market risk premium, λ_t, and the return's β_t is zero, on account of a constant conditional beta, or the market risk premium moving in proportion to the market factor's conditional variance, etc., then the simplified expected-return beta representation holds. The unconditional CAPM on this account is a specific case of a more general conditional model, and hence it is appropriate to test CAPM with a time-varying market factor loading. Note, even with a time-varying beta, a time-series regression test should produce an intercept that is statistically zero.

CAPM is a useful vehicle to generate simulated returns for the express purpose of recovering time-varying parameters because alpha and beta can take the form of any stochastic process[10].

[10] The empirical failure of the unconditional CAPM's to explain the cross-section of returns, and the documented failures of the conditional CAPM, e.g. Lewellen and Nagel (2006), are moot. For simplicity, I don't use models with more tradeable factors.

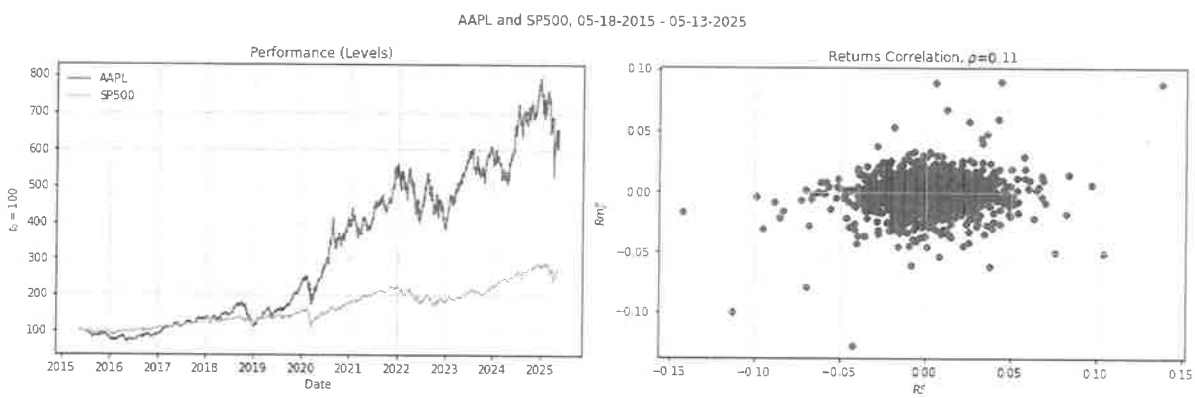
Simulation Exercise

Simulation requires estimates for the variance of the time-varying CAPM's error, the variances of α_t, β_t's error terms, and the drift and diffusion parameters of the GBM process for return simulation. The frequency of simulated returns and parameters must match the frequency of the data used to calibrate the simulation processes. Time series estimators have favorable asymptotic properties for large T, hence I use daily stock return, risk free rate, and market return data. 10 years of daily data produces T = 2,520.

The choice of stock and index should align with the premise of time-varying parameters. A stock that exhibits persistently perfect or zero co-movement with an index is unhelpful. Apple is one of the largest public corporations based upon market capitalization, and therefore it's stock performance bears strongly on that of market cap weighted indices. I use the last 10 years of AAPL and SP500 returns, spanning 5/18/2015 – 5/13/2025, and the yield on the 1-month UST bill as the risk-free rate, to calibrate parameters of the GBM process. All returns are logarithmic. UST yields are annualized and reported in percentage points, so I scale appropriately:

$$R_t = \ln(P_t^{\text{AAPL}}) - \ln(P_{t-1}^{\text{AAPL}})$$
$$Rm_t = \ln(P_t^{\text{SP500}}) - \ln(P_{t-1}^{\text{SP500}})$$
$$Rf_t = \frac{1}{365}\ln(1 + 0.01 \cdot Y_t)$$

AAPL outperforms the market over the sample period, showing stronger co-movement pre-2019 and in stress periods, such as Covid, but otherwise modest return correlation of 11% over the entire sample.



I estimate CAPM using the entire sample and constant parameters with OLS. The excess market return explains 1.2% of the variation in excess AAPL return's, which is to be expected given the noise in daily return data, but parameter estimates are significant at a 5% level using Newey-West standard errors.

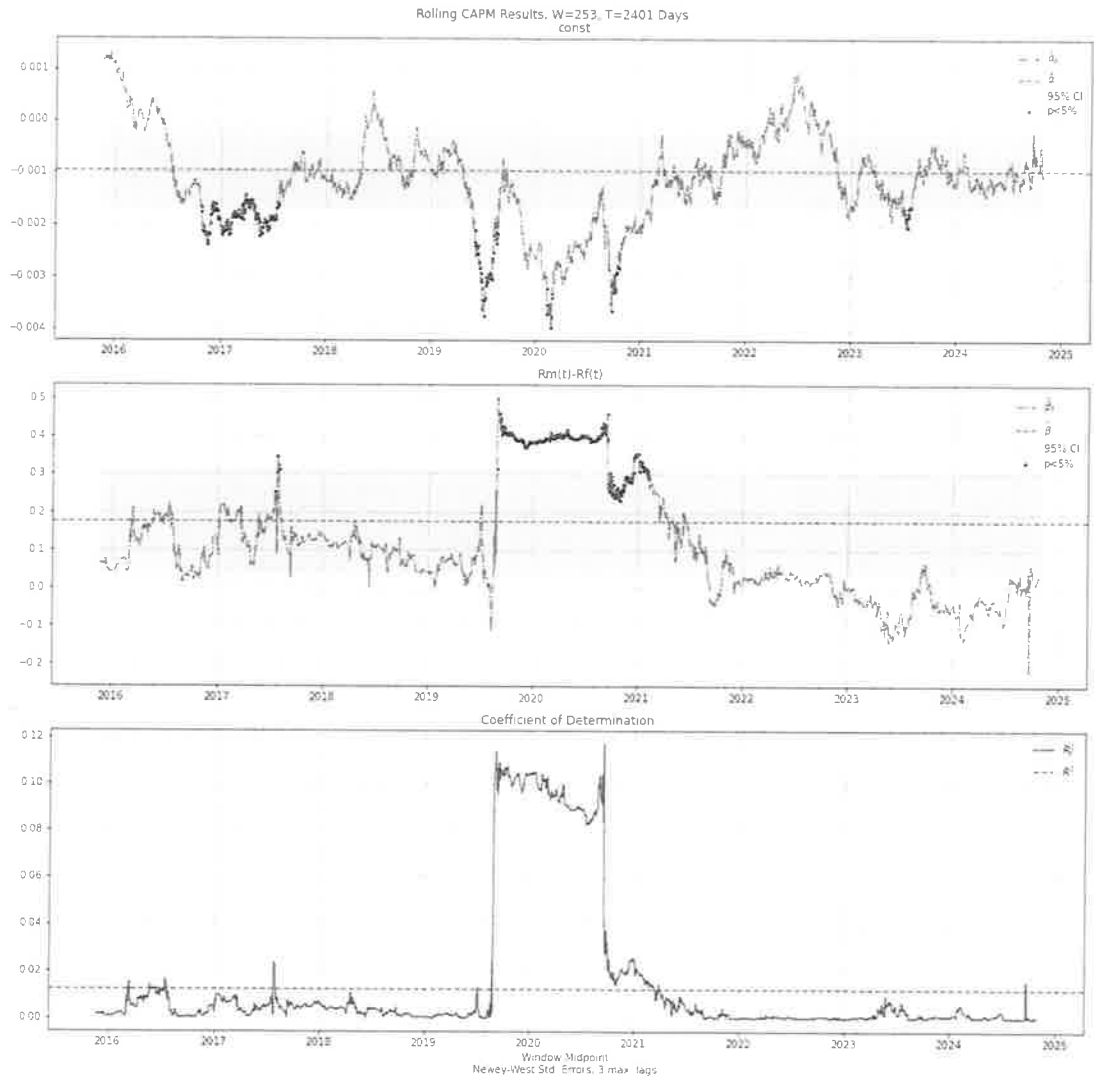
Constant Parameter Regression Results

	Coefficient	P-Value
$\hat{\alpha}$	-0.0010 (0.000)	0.012
$\hat{\beta}$	0.1736 (0.070)	0.014
$\hat{\sigma}^2_e = 0.00034$, $R^2 = 1.2\%$, T=2,401		

Results for $R^{AAPL}_t - Rf_t = \alpha + \beta(R^{SP500}_t - Rf_t) + e_t$. Newey-West standard errors in parenthesis. T=2401 daily returns (5/18/2015 – 5/13/2025).

Beta is 17.4%, and alpha is in effect zero. I perform a CUSUM test on the residuals at a 5% level and fail to reject the null hypothesis of no structural break. Parameters estimates are stable. I perform rolling OLS, using a window of size W=253 and increments of 1 day, on the same data. The result is T-W+1=2401-253+1=2149 sets of parameter estimates and associated confidence intervals, p-values, and R^2 values.

Rolling parameter estimates using a 1-year window are mostly within the 95% confidence interval of the whole sample estimates, aside from 2017 and 2020-2021. The time index in the plots below shows the midpoint of the rolling window. Model explanatory power and the frequency of statistically significant estimates peaks during the Covid stress period, and beta estimates post-2021 are lower than pre-2020.



Despite stable whole sample estimates of the constant parameter model, there is clearly at least one time-varying aspect of the relationship between the excess market return and AAPL stock (the deviation in 2020-2021). The AAPL and SP500 regression is a useful starting point to seed the simulations.

Rolling regressions produce vectors of estimated $\hat{\alpha}_t$, $\hat{\beta}_t$, dimension 2149. I perform AR(1) regressions, using 1 period lagged estimates as regressors, to sanity check the random walk formulation and estimate the variance of the residuals. The purpose is estimation for the simulation, not inference[11].

$$\hat{\alpha}_t = a_0 + \phi_a \hat{\alpha}_{t-1} + \eta_{a,t}, \eta_{a,t} \sim \mathrm{WN}(0, \sigma_a^2)$$
$$\hat{\beta}_t = b_0 + \phi_b \hat{\beta}_{t-1} + \eta_{b,t}, \eta_{b,t} \sim \mathrm{WN}(0, \sigma_b^2)$$

The AR(1) coefficients are nearly 1, and constants are zero. However, the correlation between the alpha and beta AR(1) regression residuals is -18.73%. The assumption of mutually uncorrelated error terms for the parameter random walk processes is simplifying, but the random walk formulation is reasonable.

AR(1) Regression Results

Estimate	AR(1) Model	
	$\hat{\alpha}_t$	$\hat{\beta}_t$
$\hat{a}_0$	-0.0000 (0.0000)	-
$\hat{\Phi}_a$	0.9916 (0.0025)	
$\hat{b}_0$	-	0.0008 (0.0005)
$\hat{\Phi}_b$		0.9920 (0.0027)
$\hat{\sigma}_a$	0.0001	-
$\hat{\sigma}_b$	-	0.0177
R^2	98.64%	98.40%

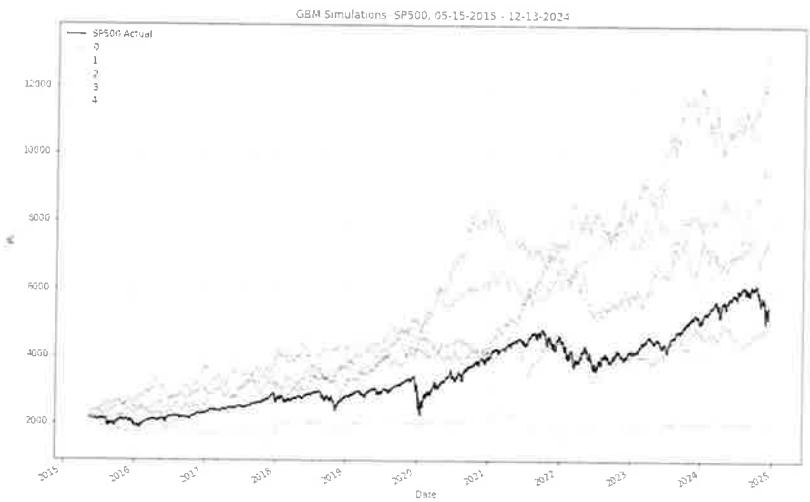
Results for regressions of T=2148 rolling OLS parameter estimates on lagged rolling OLS parameter estimates. Standard errors in parenthesis.

Calibrating the drift and diffusion parameters of the GBM process is straightforward. Per Lo and MacKinlay (1996), the maximum likelihood estimators, letting h=1 corresponding to intervals of 1, are:

$$a \equiv \mu - 0.5\sigma^2$$
$$\hat{a}_{MLE} = \frac{1}{T}\sum_{t=1}^{T} R_t$$
$$\hat{\sigma}^2_{MLE} = \frac{1}{T}\sum_{t=1}^{T}(R_t - \hat{a}_{MLE})^2$$

I compute the sample mean of daily SP500 log returns and solve for the drift and diffusion. Using T=2,420 observations spanning 5/18/2015 -5/14/2025, I obtain estimates of 0.000533 and 0.000136, respectively. Below are sample paths for the simulated market returns. To compute excess market returns, I sample nonparametrically from observed daily log returns on the 1-month US Treasury bill.

[11] There are several econometric issues. First, the regressors are generated. Second, OLS estimates of the AR(1) parameter exhibit bias, although this is primarily an issue for small T < 100. Third, if the null hypothesis of a random walk is true, the regressions are spurious.



Final parameters for the simulation process are below. I generate N=1,000 simulations of length T=2,500, using 20% burn in and mean-zero multivariate normal draws for the error terms. Aside from the GBM's S_0, which is the actual initial value of the SP500 on 5/18/2015, initial values are random.

Simulation Parameter Values

Parameter	Estimate	Calibration Details
$\hat{\mu}_{GBM}$	0.0005326406784490215	Daily SP500 log returns, 5/18/2025 – 5/14/2025 (T=2,420)
$\hat{\sigma}^2_{GBM}$	0.00013595490219255114	
$\hat{\sigma}_a$	0.0001	Sample standard deviation of AR(1) regression of T=2148 OLS parameter estimates, parameter estimates obtained using rolling OLS on T=2401 daily AAPL and SP500 excess log returns (5/18/2015 – 5/13/2025) with W=253.
$\hat{\sigma}_b$	0.0177	
$\hat{\sigma}^2_e$	0.00034	Sample variance of CAPM OLS on T=2401 daily AAPL and SP500 excess log returns (5/18/2015 – 5/13/2025)
S_0	2122.73	SP500 log return on 5/15/2015.

CV Procedures for Kernel Regression Bandwidth

I use the heuristic $h=\sqrt{T}$ in the simulation study because repeated CV with T=2,500 and N=1,000 is computationally prohibitive. However, there are several procedures to select rolling OLS' optimal window size W, and kernel regression's optimal bandwidth, h:

1. 1-step ahead forecast: for each candidate W, perform rolling OLS and use the in-sample fit to compute the one period ahead forecast. Compute the forecast error and time average of forecast errors, and select the average forecast error minimizing W. This technique extends to parameter stability. If parameters are constant, the forecast error from rolling regressions should average to 0.

2. LOOCV: for each t, omit the observation at t and fit kernel regressing using a candidate h value at t-k,...,t-1,t+1,...,t+k. At each t, compute forecast error using the forecast on the omitted t. Average the errors or take a weighted average using the kernel. Select the average forecast error minimizing h.

3. For each candidate h, perform kernel regression and obtain time-varying estimates of a_t and b_t. Fit AR(1) regression models to the estimates, and compute the sum of squared residuals. Sum the SSR for a_t and SSR for b_t to obtain a score for candidate h. Select the score minimizing h. In one regard, it's akin to picking the h that selects the best recovery of a_t and b_t, given the true processes are random walks, instead of the best recovery of y_t. However, it can be interpreted as merely selecting the test error minimizing h, given the objective is parameter recovery. An extension is to perform leave-one out estimation first, meaning for each t, omit the observation at t, obtain kernel estimates of a_t and b_t by averaging estimates a_{t-1} and a_{t+1} and b_{t-1} and b_{t+1}, and then perform the SSR procedure as described. This prevents CV from selecting a tiny bandwidth that merely tracks noise e_t in the process for y_t.

Sequential OLS with Downweighting

The crux of the challenge with time-varying parameter estimation is over what temporal subsample of the data do the parameters vary. I outline an approach grounded in the properties of a stationary AR(1) process to address this challenge. First, I detail the long run expected values of parameters governed by AR(1) processes, and draw the connection to sequential OLS. Second, I revise sequential OLS to incorporate downweighing of long dated observations using weighted least squares.

Assume a contemporaneous time series regression model has one covariate and time-varying parameters, where the parameters follow AR(1) processes:

$$\begin{aligned} Y_t &= a_t + b_t X_t + \epsilon_t & e_t &\sim \mathrm{WN}(0, \sigma_e^2) \\ a_t &= \mu_a + \phi_a a_{t-1} + w_{t,a} & w_{t,a} &\sim \mathrm{WN}(0, \sigma_a^2) \\ b_t &= \mu_b + \phi_b b_{t-1} + w_{t,b} & w_{t,b} &\sim \mathrm{WN}(0, \sigma_b^2) \end{aligned}$$

White noise terms are mutually uncorrelated. Assume further that the AR(1) coefficients ϕ have absolute value less than 1. Using recursive substitution:

$$\begin{aligned} a_t &= \mu_a + \phi_a \left(\mu_a + \phi_a a_{t-2} + w_{t-1,a}\right) + w_{t,a} \\ &= \mu_a + \phi_a \mu_a + \phi_a^2 \left(\mu_a + \phi_a a_{t-3} + w_{t-2,a}\right) + \phi_a w_{t-1,a} + w_{t,a} \\ &= \mu_a + \phi_a \mu_a + \phi_a^2 \mu_a + \phi_a^3 \left(\mu_a + \phi_a a_{t-4} + w_{t-3,a}\right) + \phi_a^2 w_{t-2,a} + \phi_a w_{t-1,a} + w_{t,a} \\ &\vdots \\ &= \mu_a \left(\sum_{j=0}^{t-1} \phi_a^j\right) + \phi_a^t a_0 + \sum_{j=0}^{t-1} \phi_a^j w_{t-j,a} \\ b_t &= \mu_b \left(\sum_{j=0}^{t-1} \phi_b^j\right) + \phi_b^t b_0 + \sum_{j=0}^{t-1} \phi_b^j w_{t-j,b} \end{aligned}$$

Substitute into Y_t and rearrange:

$$\begin{aligned} Y_t &= a_t + b_t X_t + \epsilon_t \\ &= \left[\mu_a \left(\sum_{j=0}^{t-1} \phi_a^j\right) + \phi_a^t a_0 + \sum_{j=0}^{t-1} \phi_a^j w_{t-j,a}\right] + \left[\mu_b \left(\sum_{j=0}^{t-1} \phi_b^j\right) + \phi_b^t b_0 + \sum_{j=0}^{t-1} \phi_b^j w_{t-j,b}\right] X_t + \epsilon_t \\ &= \left[\mu_a \left(\sum_{j=0}^{t-1} \phi_a^j\right) + \phi_a^t a_0\right] + \left[\mu_b \left(\sum_{j=0}^{t-1} \phi_b^j\right) + \phi_b^t b_0 + \sum_{j=0}^{t-1} \phi_b^j w_{t-j,b}\right] X_t + \left(\epsilon_t + \sum_{j=0}^{t-1} \phi_a^j w_{t-j,a}\right) \\ &= \left[\frac{\mu_a(1-\phi_a^t)}{1-\phi_a} + \phi_a^t a_0\right] + \left[\frac{\mu_b(1-\phi_b^t)}{1-\phi_b} + \phi_b^t b_0 + \sum_{j=0}^{t-1} \phi_b^j w_{t-j,b}\right] X_t + \left(\epsilon_t + \sum_{j=0}^{t-1} \phi_a^j w_{t-j,a}\right) \end{aligned}$$

As t approaches infinity, and for finite X_t:

$$\begin{aligned} Y_t &= \frac{\mu_a}{1-\phi_a} + \left[\frac{\mu_b}{1-\phi_b} + \sum_{j=0}^{t-1} \phi_b^j w_{t-j,b}\right] X_t + \left(\epsilon_t + \sum_{j=0}^{t-1} \phi_a^j w_{t-j,a}\right) \\ E(Y_t|X_t) &= \frac{\mu_a}{1-\phi_a} + \frac{\mu_b}{1-\phi_b} X_t + \sum_{j=0}^{t-1} \phi_b^j E(w_{t-j,b}|X_t) + E\left(\epsilon_t + \sum_{j=0}^{t-1} \phi_a^j w_{t-j,a}\right) \\ &= \frac{\mu_a}{1-\phi_a} + \frac{\mu_b}{1-\phi_b} X_t \end{aligned}$$

The conditional expectation of Y_t is a linear combination of X_t and the long run expected values of the coefficient's autoregressive processes. Sequential OLS expands the window over which the regression is fit incrementally, with each new time series observation, until the parameter estimates converge. If the assumption that the time-varying parameters are stationary AR(1)'s holds, it suffices to use OLS conventionally on the entire sample for large T because a_t and b_t should converge to their long run

means. Sequential OLS produces 'better' estimates of a_t and b_t for no other reason than they converge to constants. This is not the case if a_t and b_t are random walks, hence I don't use sequential OLS as an approach to compare kernel regression to state space modelling in the simulation study.

For small T, sequential OLS can perform poorly. I introduce a mechanism to down weight older observations using weighted least squares, and a regularization term in the loss function to prevent overfitting. The shrinkage parameter is time-varying, and decreases in magnitude as t increases: for small T, regularization is stronger. Estimation at t uses all prior observations but with decaying weights.

I introduce new notation for convenience. For t=1...T, let:

- y_t be a scalar response
- Y_t a t x 1 vector of responses including $y_1 \ldots y_t$
- X_t a t x k matrix of covariates
- β_t a k x 1 vector of parameters

Assume a linear relationship, for $\varepsilon_t \sim WN(0, \sigma^2_e I_t)$. W_t is a diagonal matrix with elements $\lambda^{t-1}, \lambda^{t-2}, \ldots, \lambda^{t-(t-1)}$, 1. The dimensions of W_t naturally vary with t:

$$W_t = \begin{bmatrix} \lambda^{t-1} & \cdots & & 0 \\ & \lambda^{t-2} & & \\ \vdots & & \ddots & \vdots \\ & & \lambda & \\ 0 & & & 1 \end{bmatrix}_{t \times t}$$

As t increases for fraction λ, weights decay to 0. Let $w_t = W_t^{0.5}$, and note that $w_t' w_t = W_t$. For estimation purposes, iterate from t = 3...T, and minimize the weighted loss function:

$$\begin{aligned} L_t(\beta_t) &= ||w_t \epsilon_t||_2^2 + \alpha_t ||\beta_t||_2^2 \\ &= (w_t \epsilon_t)' (w_t \epsilon_t) + \alpha_t \beta_t' \beta_t \\ &= \epsilon_t' w_t' w_t \epsilon_t + \alpha_t \beta_t' \beta_t \\ &= (Y_t - X_t \beta_t)' W_t (Y_t - X_t \beta_t) + \alpha_t \beta_t' \beta_t \end{aligned}$$

where α_t is a time-varying shrinkage parameter such that regularization strength decreases as t approaches T. Solving:

$$\begin{aligned} \min_\beta L_t(\beta) &= (Y_t - X_t \beta)' W_t (Y_t - X_t \beta) + \alpha_t \beta' \beta \\ \nabla_\beta L_t(\beta) &= 0 \\ 0 &= \nabla_\beta (Y_t' W_t - \beta' X_t' W_t)(Y_t - X_t \beta) + \alpha_t \beta' \beta \\ &= \nabla_\beta \left(Y_t' W_t Y_t - \beta' X_t' W_t Y_t - Y_t' W_t X_t \beta + \beta' X_t' W_t X_t \beta + \alpha_t \beta' \beta \right) \\ 0 &= -X_t' W_t Y_t - X_t' W_t' Y_t + 2 X_t' W_t X_t \beta + 2 \alpha_t I_t \beta \\ 2 X_t' W_t Y_t &= 2 (X_t' W_t X_t + \alpha_t I_t) \beta \\ \beta^* &= (X_t' W_t X_t + \alpha_t I_t)^{-1} \left(X_t' W_t Y_t \right) \end{aligned}$$

As expected, at $W_t = \sigma^2_e I_t$ and $\alpha_t = 0$, the estimator collapses to the OLS estimator. The simplest form for α_t is a linear time trend:

$$\alpha_t = a_0 + a_1 (T - t), a_0, a_1 \geq 0$$

but exponential or more complicated functions are possible. The estimator's variance, and therefore standard errors, confidence intervals, and p-values follow from:

$$
\begin{aligned}
\beta^* &= (X_t' W_t X_t + \alpha_t I_t)^{-1} \left(X_t' W_t Y_t \right) \\
\mathrm{Var}(\beta^* | X_t) &= (X_t' W_t X_t + \alpha_t I_t)^{-1} \mathrm{Var} \left(X_t' W_t X_t \beta_t + X_t' W_t \epsilon_t | X_t \right) (X_t' W_t X_t + \alpha_t I_t)^{-1} \\
&= (X_t' W_t X_t + \alpha_t I_t)^{-1} X_t' W_t \sigma_e^2 I_t W_t' X_t (X_t' W_t X_t + \alpha_t I_t)^{-1} \\
&= \sigma_e^2 (X_t' W_t X_t + \alpha_t I_t)^{-1} X_t' W_t W_t' X_t (X_t' W_t X_t + \alpha_t I_t)^{-1}
\end{aligned}
$$

Thus far I've assumed homoscedastic errors. To estimate variance σ^2_e, normalize the weighted sum of squared errors by the degrees of freedom:

$$
\begin{aligned}
\hat{\sigma}^2_{e,t} &= \frac{(Y_t - X_t \hat{\beta}_t)' W_t (Y_t - X_t \hat{\beta}_t)}{t - 2 - 1} \\
\widehat{\mathrm{Cov}}(\beta_t) &= \hat{\sigma}^2_{e,t} (X_t' W_t X_t + a_t I_t)^{-1} X_t' W_t' W_t X_t (X_t' W_t X_t + a_t I_t)^{-1}
\end{aligned}
$$

95% confidence intervals take the form:

$$
\begin{aligned}
\hat{\beta}_{0,t} &\pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_{0,t}) \\
\hat{\beta}_{1,t} &\pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_{1,t})
\end{aligned}
$$

As the subscripts indicate, sequential OLS with downweighing and regularization produces estimates, confidence intervals, p-values, and measures of goodness of fit (e.g. R^2, MSE), at each time index t.

Rolling OLS

Algorithmically, rolling OLS proceeds as follows:

Assuming the increment between consecutive rolling windows is 1

For t in 1....T:

Take a subset of size W+1 of the dependent variable and covariates

$$Y_t = [y_{t-\frac{w}{2}} \dots y_t \dots y_{t+\frac{w}{2}}]'$$

$$X_t = [x_{t-\frac{w}{2}} \dots x_t \dots x_{t+\frac{w}{2}}]'$$

Solve for β:

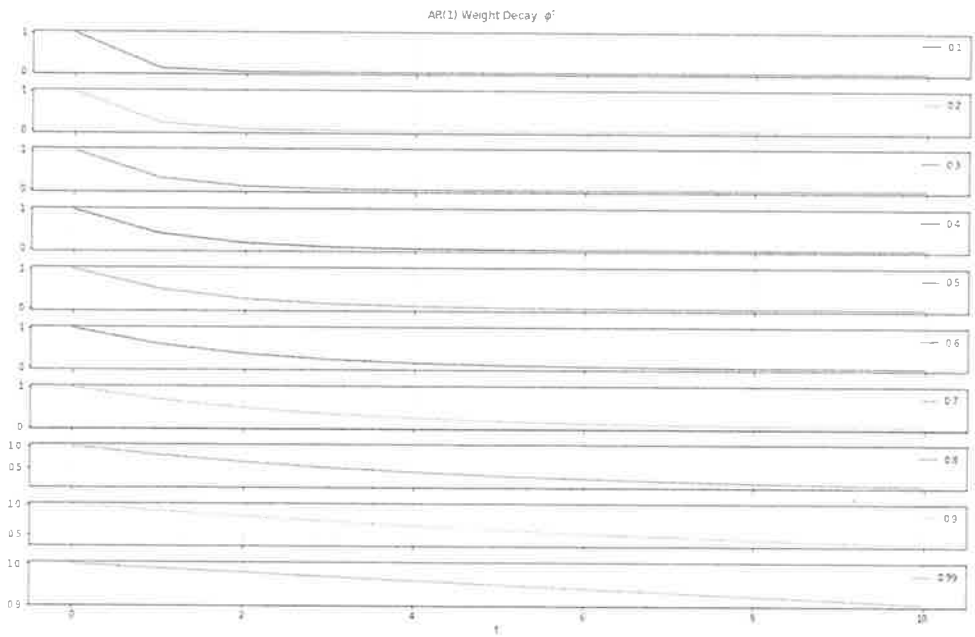
$$\min_\beta ||(Y_t - X_t\beta)||_2^2$$

Collect $\hat{\beta}_t$ and associated measure of statistical significance and goodness of fit

The only hyperparameter is the window size, W. W should be large enough to provide sufficient statistical power for the regression, but small enough to identify variation in the parameters. Larger windows produce smoother estimates than shorter windows. The optimal window size should be informed by the domain but can be selected using cross-validation.

One-Sided ACF Window

A Gaussian kernel applied to observations at t_0 uses strictly positive weights for observations $t > t_0$, and therefore is a two-sided window function. It may be prudent to use a one-sided function, for at least two reasons. First, if the parameters of the data generating process are random walks, and the covariate is an IID draw, observations of Y(t) at $t > t_0$ shouldn't systematically inform estimation of parameters a_t and b_t. Second, prediction of a_{t+1} and b_{t+1} can apply non-negative weights to observations $t \leq t_0$ but must zero weights to observations $t > t_0$. Such a function is not symmetric, and therefore not a kernel. An obvious candidate function is the ACF, which captures the persistence of the process and therefore the relative importance of observations at t close to t_0. For an AR(1), the ACF is ϕ^h, which decays to 0 at a speed contingent on ϕ's magnitude:



The case of ϕ=0.99 is a near random walk process. The weighting function captures the strong persistence of shocks, showing weights between 0.90 and 1.0 for t_0 to t_{-10}. Using the function

$$K(x) = \mathbf{1}(x \geq 0) \cdot |\phi|^x$$

for x = t-i, t the time index at which to estimate parameters, and index i=1...T, evaluates to 0 when t < i. The more temporally prior observation i is to time t, the lower the weight assigned, for $0 < |\phi| < 1$.

I repeat the kernel regression procedure using the ACF window function, calibrating ϕ with an AR(1) regression of simulated Y_t on Y_{t-1}.

Euler Discretization

A GBM is a continuous time stochastic process. In practice, simulation requires discretizing the process by approximating the stochastic differential equation for minute time steps. I provide the details below:

For a general SDE, with drift $a(X_t, t)$ and diffusion $b(X_t, t)$, and using the properties of a Brownian motion (increments over non-overlapping time intervals are independent and normally distributed):

$$\begin{aligned} dX_t &= a(X_t, t)dt + b(X_t, t)dW_t \\ \Delta W_t &= W_{t+\Delta t} - W_t \\ &\sim N(0, \sqrt{\Delta t}^2) \\ \Delta W_t &= \sqrt{\Delta t}Z \\ Z &\sim N(0, 1) \end{aligned}$$

Integrate the SDE:

$$\begin{aligned} \int_t^{t+\Delta t} dX_s &= \int_t^{t+\Delta t} a(X_s, s)ds + \int_t^{t+\Delta t} b(X_s, s)dW_s \\ X_{t+\Delta t} - X_t &= \int_t^{t+\Delta t} a(X_s, s)ds + \int_t^{t+\Delta t} b(X_s, s)dW_s \end{aligned}$$

Approximate the integral by assuming that for a minute increment Δt, the drift and diffusion terms are approximately constant:

$$\begin{aligned} \int_t^{t+\Delta t} a(X_s, s)ds &\approx a(X_t, t)\int_t^{t+\Delta t} ds \\ &= a(X_t, t)\Delta t \\ \int_t^{t+\Delta t} b(X_s, s)dW_s &\approx b(X_t, t)\int_t^{t+\Delta t} dW_s \\ &= b(X_t, t)(W_{t+\Delta t} - W_t) \\ &= b(X_t, t)\Delta W_t \end{aligned}$$

Substitute and rearrange:

$$\begin{aligned} X_{t+\Delta t} - X_t &\approx a(X_t, t)\Delta t + b(X_t, t)\Delta W_t \\ &= X_t + a(X_t, t)\Delta t + b(X_t, t)\Delta W_t \end{aligned}$$

For a GBM,

$$\begin{aligned} dS_t &= \mu S_t dt + \sigma S_t dW_t^P \\ S_{t+\Delta t} &\approx S_t + \mu S_t \Delta t + \sigma S_t \Delta W_t^P \\ &= S_t + \mu S_t \Delta t + \sigma S_t \sqrt{\Delta t}Z \end{aligned}$$

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Attachments

Code:

Time_Varying_Parameters_PT1.ipynb

Time_Varying_Parameters_PT1.html

Time_Varying_Parameters_PT1.ipynb

Time_Varying_Parameters_PT2.html

Simulation Data:

SP500.csv

DGS1MO.csv

HistoricalData_1747318668456.csv

Simulation Results:

kalman_filter_results.csv

kernel_test_results.csv